Authoritative technical references

Research baseline: **2026-11-05 (America/Toronto)**. These are primary-source references for the proposed design, not a claim that referenced software is installed. Recheck supported versions, advisories, license/plan terms, commands, and hardware compatibility before each approved stage. Source fetch availability and version-specific ambiguity must be reported rather than guessed.

1. Standards and namespaces

- NIST SP 800-207: Zero Trust Architecture - resource-focused principles and logical architecture.
- NIST SP 800-53 Revision 5 - security/privacy control catalog; use the current release material linked there.
- NIST SP 800-53 Release 5.2.0 notice - release baseline context, not a new revision number.
- CIS Controls and CIS Controls v8.1 - prioritized control areas; project mappings are tailored educational interpretations.
- RFC 2606: Reserved Top Level DNS Names - `.test` for testing; no corporate domain required.
- RFC 6761: Special-Use Domain Names - special-use name behavior and considerations.

2. Proxmox

- Official documentation downloads - choose the version-matched administration guide at implementation time.
- System requirements - virtualization and host requirements; supplied RAM does not establish compatibility.
- Network configuration - bridge/interface concepts; actual interface names must be inspected.
- Installation and upgrade - supported installation paths and release-specific cautions.
- Administration guide PDF - management, firewall, backup, and recovery reference. Some Proxmox documentation endpoints restrict automated access; use the official download index if a direct fetch is refused.

3. OPNsense

- Virtual and cloud installation - VM deployment entry point.
- Firewall rules and processing - stateful policy, rule ordering, and default behavior.
- Interface configuration - interface assignment and private/bogon filtering implications.
- Official documentation - current version-specific backup, logging, and hardening procedures.

4. Windows evaluation and hardware

- Windows Server 2025 Evaluation Center - evaluation media, 180-day expiry, and stated initial activation requirement. Recheck terms and deadlines at use; no license entitlement or purchase is implied.
- Windows Server hardware requirements - CPU, memory, and storage prerequisites; do not treat minimums as measured workload sizing.

5. Readiness and network commands

- Get-CimInstance - local read-only CIM inventory syntax used in Stage A.
- Get-NetIPConfiguration - local interface/DNS configuration inspection; output is private.

6. Active Directory, DNS, and policy

- Install Active Directory Domain Services - supported AD DS role installation and deployment workflow.
- Install-WindowsFeature - verified role-installation syntax; changes the approved target.
- Install-ADDSForest - verified new-forest/DNS syntax, secure DSRM prompt, reboot/functional-level considerations.
- AD domain firewall requirements - service dependencies and dynamic RPC scope; does not authorize production trust.
- Core network components - lab DNS/directory network deployment context.

7. Endpoint and certificate verification

- Test-ComputerSecureChannel - member-machine secure-channel check; not a DC health test.
- AD CS overview - PKI service components.
- Certification Authority role - standalone/offline root and subordinate CA concepts.
- Inspect certificates and stores - verified certificate inspection and `certutil -verify -urlfetch` syntax; fetch URLs must be reviewed.
- Certificate path validation and revocation policy - relying-party trust/revocation configuration; validation tools do not establish application behavior by themselves.

8. Tailscale

- Access control - policy model and enforcement.
- Default ACL and policy examples - initial allow-all warning; do not infer deny-all from missing policy.
- Grant examples - explicit resource-scoped grants and additive access.
- Device visibility - identity/environment membership and default visibility concerns.
- Exit nodes - capability deliberately excluded from this baseline.
- Tailscale CLI - verify `tailscale status` against the installed version; output contains private identifiers.

9. Twingate

- Understanding connectors - outbound connector/resource model.
- Resources - exact resource addressing, DNS perspective, and explicit port restrictions; defaults require review.
- Connectors - supported deployment methods; no token-bearing deployment command is copied into Git.
- How firewalls work with Twingate - resource scope, network controls, and audit considerations.
- Connector connectivity troubleshooting - current outbound Controller/Relay dependencies; verify current protocols/ports rather than assume HTTPS alone is sufficient.

10. Logging and automation

- Windows Event Forwarding for detection - collection design and source coverage.
- Source-initiated event subscriptions - source permissions and subscription setup.
- systemd journalctl - verified `--since` and `--no-pager` inspection options; use the installed distribution's version.
- Ansible check and diff modes - simulation/module limitations and sensitive diff output.
- Ansible playbook CLI - syntax-check/check/diff/limit options; target inventory remains private.

11. GitHub and repository workflow

- GitHub Actions security hardening - least privilege and full-commit pinning.
- Official checkout action - bootstrap pins a full commit resolved from the official `v6` ref on the research date; stored checkout credentials are disabled.
- GitHub CLI repository creation - supported secure repository creation/public visibility workflow.
- GitHub CLI authentication status - inspect existing supported authentication without exposing a token.

The offline architecture makes no claim about Google Wifi VLAN capabilities and requires no home-network change. Hardware/network capability verification belongs to the separately approved readiness stage.

12. Command verification limits

Proposed Stage A/E/F/G/I/J/L examples use syntax confirmed in the corresponding primary documentation. They were not run on lab assets. Installation details, firewall backend/rules, Linux distribution commands, recovery commands, software versions, provider plan features, and actual target values intentionally require a version-specific reviewed change record before execution. Python/Git repository checks are verified locally during Task 001.